

Lancaster University College
at Beijing Jiaotong University

Digital Signal Processing

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- Frequency domain description of discrete LTI systems
- Frequency domain analysis of response of discrete LTI system
- Ideal digital filter

Frequency domain description of discrete LTI systems

Frequency domain description of discrete systems: system frequency response $H(e^{j\Omega})$

The relationship between frequency response $H(e^{j\Omega})$ and unit impulse response $h[k]$

$$H(e^{j\Omega}) = \sum_{k=-\infty}^{\infty} h[k] \cdot e^{-j\Omega k}$$

$$h[k] \xleftrightarrow{\text{DTFT}} H(e^{j\Omega})$$

Time Domain Description of Discrete LTI Systems

Frequency domain description of discrete LTI systems

Frequency domain description of discrete LTI systems

Frequency response of discrete LTI system:

$$x[k]=\mathrm{e}^{\mathrm{j}\Omega k} \longrightarrow \boxed{H(\mathrm{e}^{\mathrm{j}\Omega})} \longrightarrow y[k]=\mathrm{e}^{\mathrm{j}\Omega k}H(\mathrm{e}^{\mathrm{j}\Omega})$$

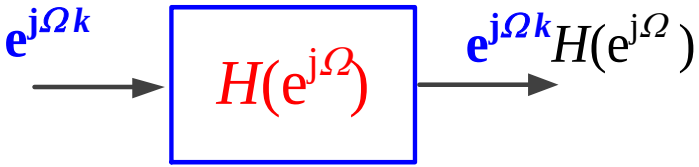
$$y[k]=\mathrm{e}^{\mathrm{j}\Omega k} * h[k]=\sum_n h[n]\mathrm{e}^{\mathrm{j}\Omega(k-n)}$$

$$=\mathrm{e}^{\mathrm{j}\Omega k}\sum_n h[n]\mathrm{e}^{-\mathrm{j}\Omega n}$$

$$=\mathrm{e}^{\mathrm{j}\Omega k}H(\mathrm{e}^{\mathrm{j}\Omega})$$

$H(\mathrm{e}^{\mathrm{j}\Omega})$: system frequency response

The physical meaning of $H(e^{j\Omega})$



The response of the imaginary exponential signal $e^{j\Omega k}$ with frequency Ω through the discrete LTI system remains the same frequency imaginary exponential signal.

$H(e^{j\Omega})$ reflects the transmission characteristics of discrete systems for imaginary exponential signals of different frequencies.

Amplitude response and phase response

$$H(e^{j\Omega}) = |H(e^{j\Omega})|\, e^{j\varphi(\Omega)}$$

amplitude response

phase response

If $h[k]$ is a real sequence, then

The amplitude response $|H(e^{j\Omega})|$ is an even function of Ω,

The phase response $j(\Omega)$ is an odd function of Ω.

[E.g] The difference equation describing a causally stable discrete LTI system is known to be $y[k] - 0.75y[k-1] + 0.125y[k-2] = 4x[k]+3x[k-1]$,
Calculate the frequency response $H(e^{j\Omega})$ and unit impulse response $h[k]$ of the system.

Solution: Performing DTFT on both sides of the difference equation yields

$$(1-0.75e^{-j\Omega}+0.125e^{-j2\Omega})Y_{zs}(e^{j\Omega})=(4+3e^{-j\Omega})X(e^{j\Omega})$$

According to the relationship between $H(e^{j\Omega})$ and $X(e^{j\Omega})$ and $Y_{zs}(e^{j\Omega})$, it can be concluded that

$$H(e^{j\Omega})=\frac{Y_{zs}(e^{j\Omega})}{X(e^{j\Omega})}=\frac{4+3e^{-j\Omega}}{1-0.75e^{-j\Omega}+0.125e^{-j2\Omega}}$$

[E.g] The difference equation describing a causally stable discrete LTI system is known to be

$y[k] - 0.75y[k-1] + 0.125y[k-2] = 4x[k]+3x[k-1]$,

Calculate the frequency response $H(e^{j\Omega})$ and unit impulse response $h[k]$ of the system.

Solution:

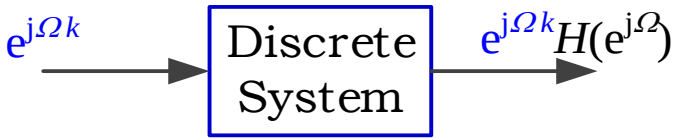
$$H(e^{j\Omega}) = \frac{Y_{zs}(e^{j\Omega})}{X(e^{j\Omega})} = \frac{4+3e^{-j\Omega}}{1-0.75e^{-j\Omega}+0.125e^{-j2\Omega}}$$

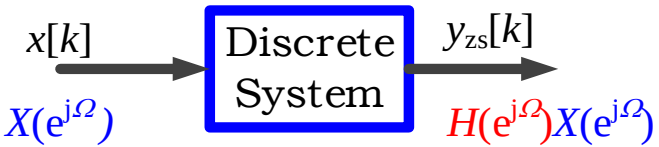
$$= \frac{20}{1-0.5e^{-j\Omega}} - \frac{16}{1-0.25e^{-j\Omega}}$$

Performing IDTFT on the above equation yields:

$$h[k] = 20\times 0.5^k u[k] - 16\times 0.25^k u[k]$$

- Response of non-periodic sequences through discrete LTI systems



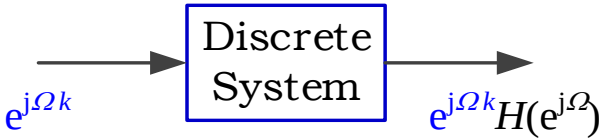


$$x[k]=\frac{1}{2\pi}\int_{-\pi}^{\pi}X(e^{j\Omega})\,e^{j\Omega k}d\Omega$$

$$y_{zs}[k]=\frac{1}{2\pi}\int_{-\pi}^{\pi}X(e^{j\Omega})H(e^{j\Omega})e^{j\Omega k}d\Omega$$

$$Y_{zs}(e^{j\Omega})=H(e^{j\Omega})X(e^{j\Omega})$$

- Response of periodic sequences through discrete LTI systems



$$\tilde{x}[k] = \frac{1}{N}\sum_{m=<N>}\tilde{X}[m]e^{j\frac{2\pi m}{N}k}$$

$$\tilde{y}[k] = \frac{1}{N}\sum_{m=<N>}\tilde{X}[m]\,H(e^{j\frac{2\pi m}{N}})e^{j\frac{2\pi m}{N}k}$$

E.g Try to calculate the response of the cosine signal $\cos(\Omega_0 k)$ through the discrete LTI system $H(\mathrm{e}^{\mathrm{j}\Omega})$.

Solution: According to Euler's formula

$$\cos(\Omega_0 k) = (\mathrm{e}^{\mathrm{j}\Omega_0 k} + \mathrm{e}^{-\mathrm{j}\Omega_0 k})/2$$

Based on the response of the virtual index signal through the system and the characteristics of the LTI system

$$y[k] = \left[H(\mathrm{e}^{\mathrm{j}\Omega_0})\mathrm{e}^{\mathrm{j}\Omega_0 k} + H(\mathrm{e}^{-\mathrm{j}\Omega_0})\mathrm{e}^{-\mathrm{j}\Omega_0 k} \right]/2$$

$h[k]$ is a real sequence $\left|H(\mathrm{e}^{\mathrm{j}\Omega_0})\right|\mathrm{e}^{\mathrm{j}\varphi(\Omega_0)}$ $\left|H(\mathrm{e}^{\mathrm{j}\Omega_0})\right|\mathrm{e}^{-\mathrm{j}\varphi(\Omega_0)}$

$$y[k] = \left|H(\mathrm{e}^{\mathrm{j}\Omega_0})\right| \cos(\Omega_0 k + \varphi(\Omega_0))$$

$$\sin(\Omega_0 k+\theta) \xrightarrow{H} \left|H(\mathrm{e}^{\mathrm{j}\Omega_0})\right| \sin\left(\Omega_0 k+\varphi(\Omega_0)+\theta\right)$$

$$\cos(\Omega_0 k+\theta) \xrightarrow{H} \left|H(\mathrm{e}^{\mathrm{j}\Omega_0})\right| \cos\left(\Omega_0 k+\varphi(\Omega_0)+\theta\right)$$

Sine type signal passing through a discrete LTI system with a frequency response of $H(\mathrm{e}^{\mathrm{j}\Omega})$

The zero state response remains a sinusoidal signal of the same frequency:

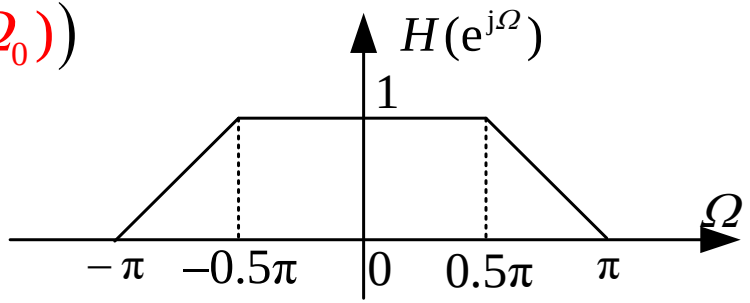
The amplitude change of the response is determined by $|H(\mathrm{e}^{\mathrm{j}\Omega})|$,

The phase change of the response is determined by $\mathrm{j}(\Omega)$.

[E.g] Given the frequency response $H(e^{j\Omega})$ of a certain discrete LTI system as shown in the figure, try to calculate the signal

$x[k] = 1+3\cos(0.5\pi k)+2\cos(0.75\pi k)+\cos(\pi k)$ through the response $y[k]$ of the system.

解：$\cos(\Omega_0 k) \xrightarrow{H} |H(e^{j\Omega_0})| \cos(\Omega_0 k + \varphi(\Omega_0))$



$$y[k] = \underbrace{|H(e^{j0})|}_{=1} \cdot 1 + \underbrace{|H(e^{j0.5\pi})|}_{=1} \cdot 3\cos(0.5\pi k)$$

$$+ \underbrace{|H(e^{j0.75\pi})|}_{=0.5} \cdot 2\cos(0.75\pi k) + \underbrace{|H(e^{j\pi})|}_{=0} \cdot \cos(\pi k)$$

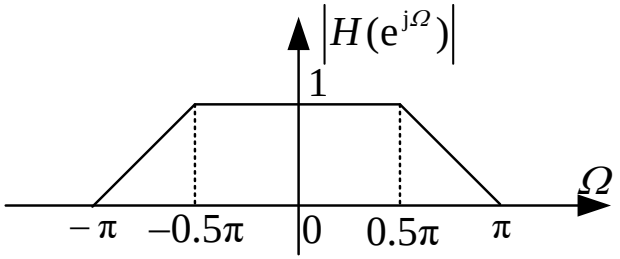
$$= 1 + 3\cos(0.5\pi k) + \cos(0.75\pi k)$$

[E.g] Given the frequency response $H(e^{j\Omega})$ of a certain discrete LTI system as shown in the figure, try to calculate the signal

$x[k] = 1+3\cos(0.5\pi k)+2\cos(0.75\pi k)+\cos(\pi k)$ through the response $y[k]$ of the system.

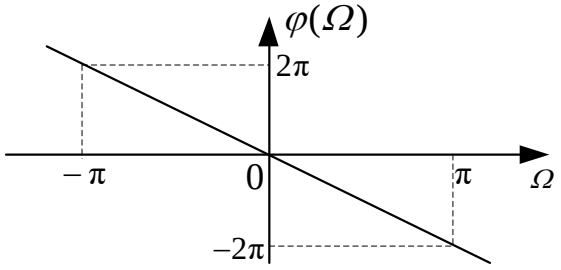
Solution:

$$\cos(\Omega_0 k) \xrightarrow{H} \left|H(e^{j\Omega_0})\right| \cos\left(\Omega_0 k + \varphi(\Omega_0)\right)$$



$$\left|H(e^{j0})\right| = 1,\ \left|H(e^{j0.5\pi})\right| = 1,\ \left|H(e^{j0.75\pi})\right| = 0.5,\ \left|H(e^{j\pi})\right| = 0$$

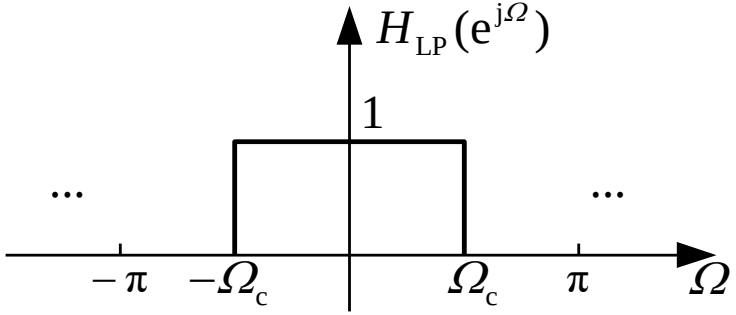
$$\varphi(0) = 0,\ \varphi(0.5\pi) = -2\times 0.5\pi,$$
$$\varphi(0.75\pi) = -2\times 0.75\pi,\ \varphi(\pi) = -2\times\pi$$



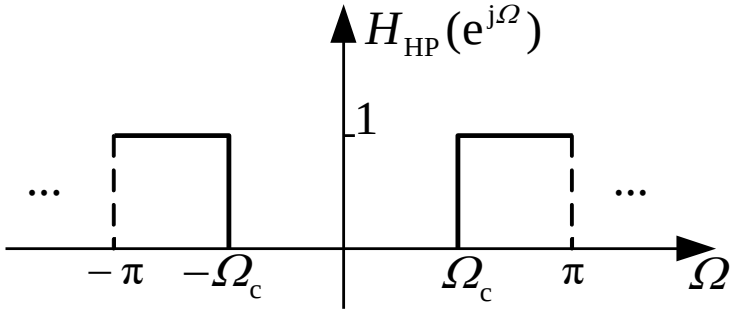
$$y[k] = \left|H(e^{j0})\right|\cdot 1 + 3\left|H(e^{j0.5\pi})\right|\cos[0.5\pi k + \varphi(0.5\pi)]$$
$$+ 2\left|H(e^{j0.75\pi})\right|\cos[0.75\pi k + \varphi(0.75\pi)] + \left|H(e^{j\pi})\right|\cos[\pi k + \varphi(\pi)]$$

$$= 1 + 3\cos[0.5\pi(k-2)] + \cos[0.75\pi(k-2)]$$

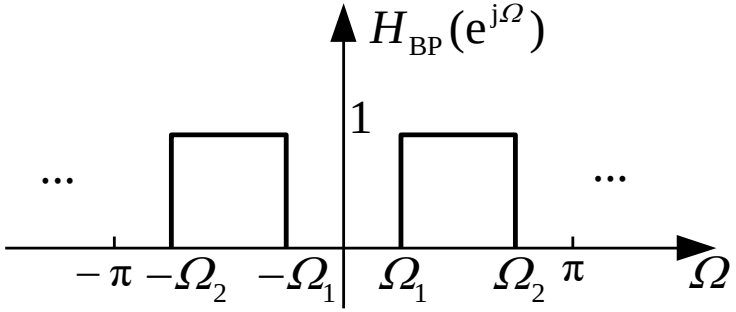
Ideal digital filter



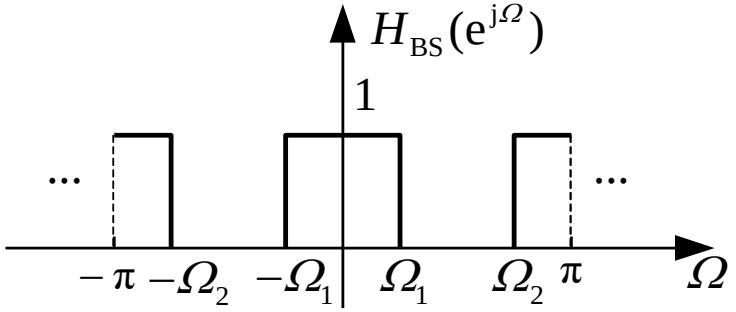
Ideal low-pass filter



Ideal high-pass filter



Ideal band-pass filter



Ideal band-stop filter

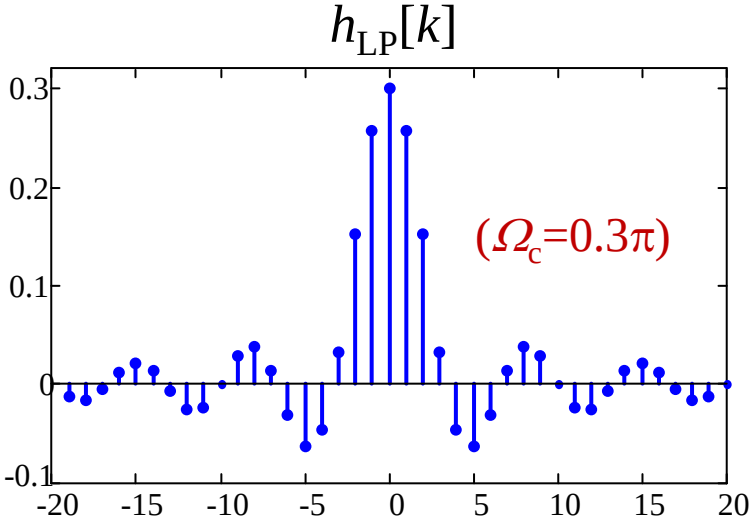
Ideal digital filter

The impulse response of an ideal low-pass digital filter

$$h_{\mathrm{LP}}[k]=\frac{1}{2\pi}\int_{-\pi}^{\pi} H_{\mathrm{LP}}(\mathrm{e}^{\mathrm{j}\Omega})\mathrm{e}^{\mathrm{j}\Omega k}\mathrm{d}\Omega$$

$$=\frac{1}{2\pi}\int_{-\Omega_{\mathrm{c}}}^{\Omega_{\mathrm{c}}} \mathrm{e}^{\mathrm{j}\Omega k}\mathrm{d}\Omega$$

$$=\frac{\Omega_{\mathrm{c}}}{\pi}\mathrm{Sa}(\Omega_{\mathrm{c}}k)$$

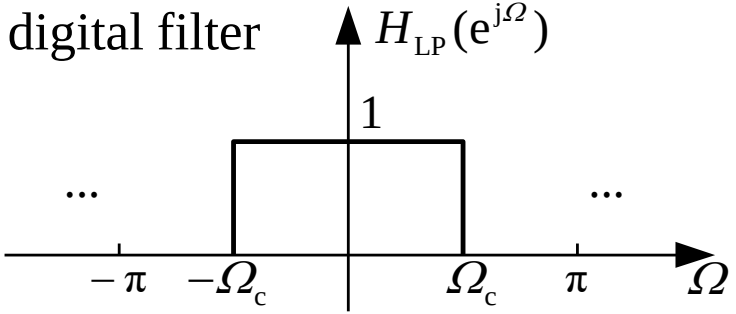


The ideal low-pass digital filter is a non- causal system.

Ideal digital filter

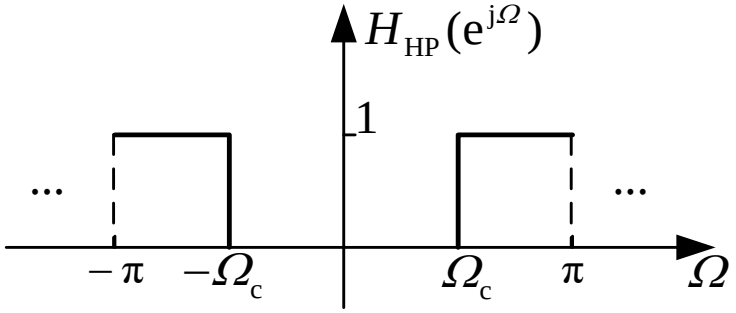
The impulse response of an ideal high pass digital filter

$$H_{\mathrm{HP}}(\mathrm{e}^{\mathrm{j}\Omega}) = 1 - H_{\mathrm{LP}}(\mathrm{e}^{\mathrm{j}\Omega})$$



$$h_{\mathrm{HP}}[k] = \delta[k] - h_{\mathrm{LP}}[k]$$

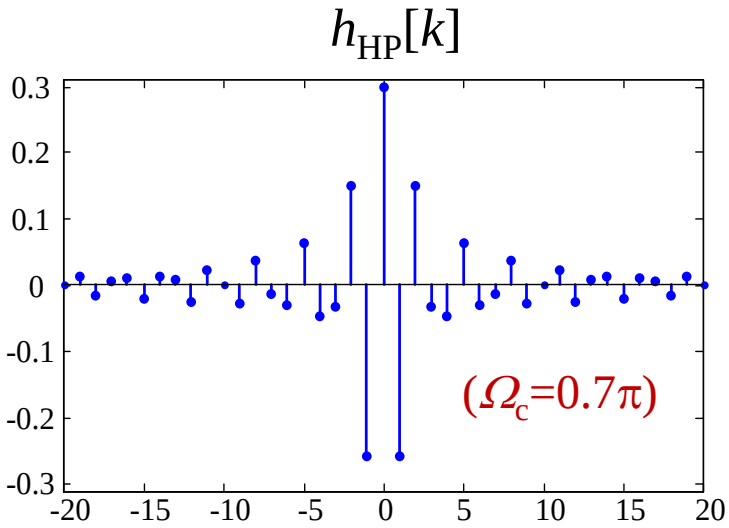
$$= \delta[k] - \frac{\Omega_c}{\pi}\mathrm{Sa}(\Omega_c k)$$



Ideal digital filter

The impulse response of an ideal high pass filter

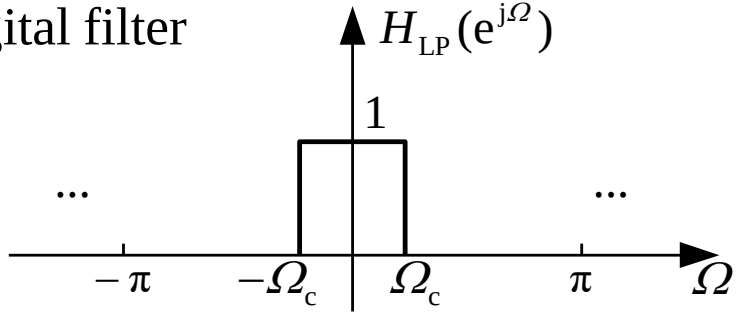
$$h_{\mathrm{HP}}[k] = \delta[k] - \frac{\Omega_{\mathrm{c}}}{\pi}\mathrm{Sa}(\Omega_{\mathrm{c}}k)$$



Ideal digital filter

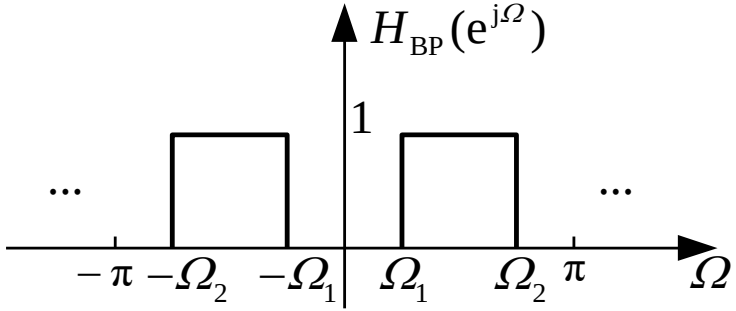
The impulse response of an ideal bandpass digital filter

$$H_{\mathrm{BP}}(\mathrm{e}^{\mathrm{j}\Omega}) = H_{\mathrm{LP2}}(\mathrm{e}^{\mathrm{j}\Omega}) - H_{\mathrm{LP1}}(\mathrm{e}^{\mathrm{j}\Omega})$$



$$h_{\mathrm{BP}}[k] = h_{\mathrm{LP2}}[k] - h_{\mathrm{LP1}}[k]$$

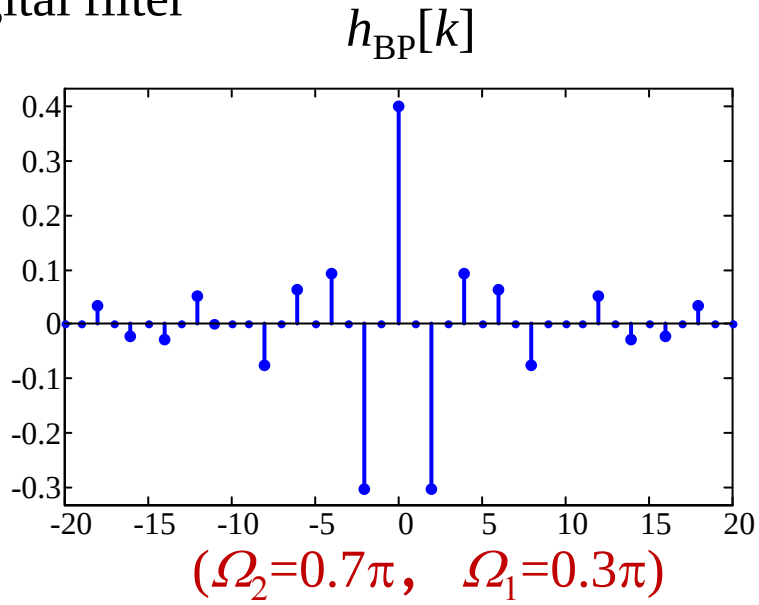
$$= \frac{\Omega_2}{\pi}\mathrm{Sa}(\Omega_2 k) - \frac{\Omega_1}{\pi}\mathrm{Sa}(\Omega_1 k)$$



Ideal digital filter

The impulse response of an ideal bandpass digital filter

$$h_{\mathrm{BP}}[k]=\frac{\Omega_2}{\pi}\mathrm{Sa}(\Omega_2 k)-\frac{\Omega_1}{\pi}\mathrm{Sa}(\Omega_1 k)$$



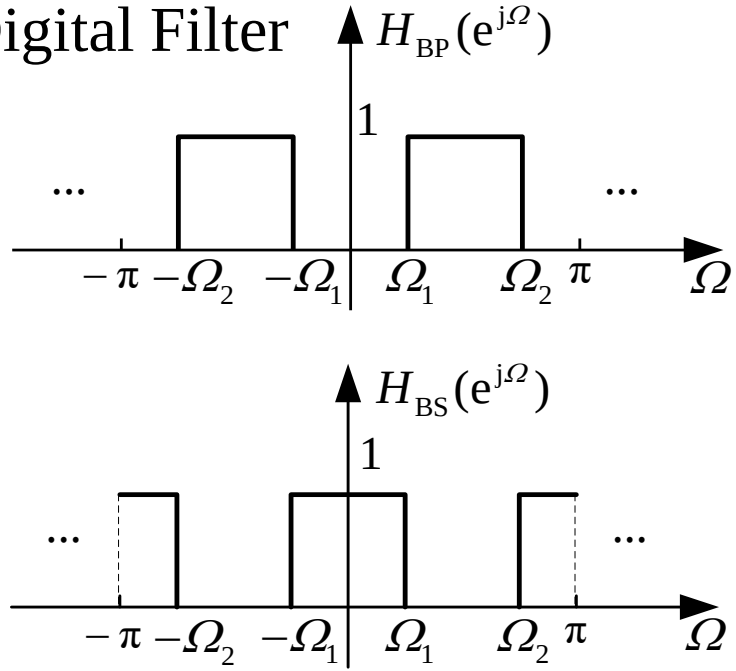
Ideal digital filter

Pulse Response of Ideal Bandstop Digital Filter

$$H_{\mathrm{BS}}(\mathrm{e}^{\mathrm{j}\Omega}) = 1 - H_{\mathrm{BP}}(\mathrm{e}^{\mathrm{j}\Omega})$$

$$h_{\mathrm{BS}}[k] = \delta[k] - h_{\mathrm{BP}}[k]$$

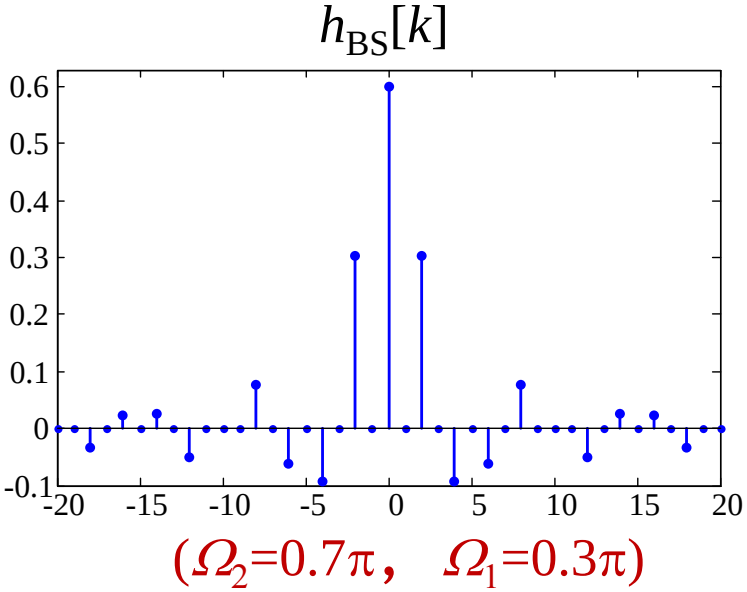
$$= \delta[k] + \frac{\Omega_1}{\pi}\mathrm{Sa}(\Omega_1 k) - \frac{\Omega_2}{\pi}\mathrm{Sa}(\Omega_2 k)$$



Ideal digital filter

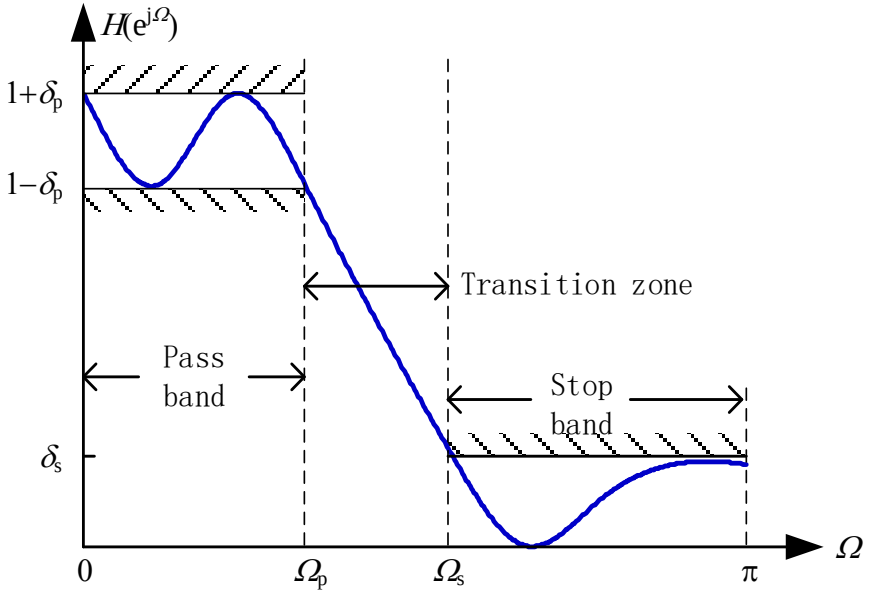
Pulse Response of Ideal Bandstop Digital Filter

$$h_{BS}[k]=\delta[k]+\frac{\Omega_1}{\pi}\mathrm{Sa}(\Omega_1 k)-\frac{\Omega_2}{\pi}\mathrm{Sa}(\Omega_2 k)$$

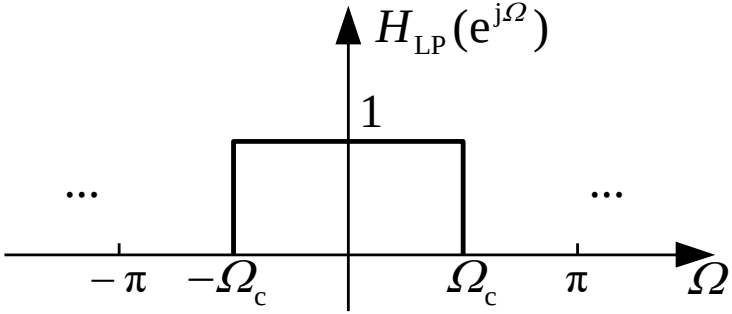


(Ω_2=0.7π， Ω_1=0.3π)

Ideal digital filter

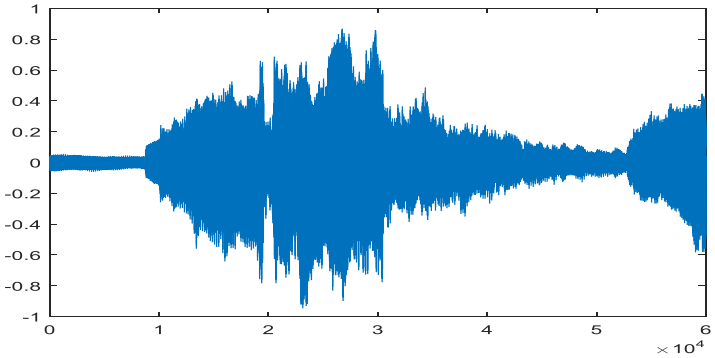


Actual low-pass digital filter

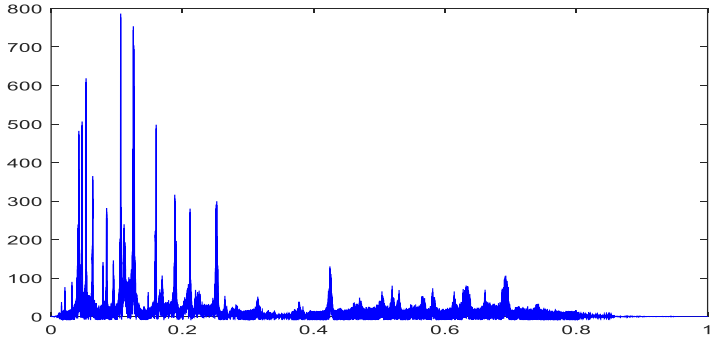


Ideal low-pass digital filter

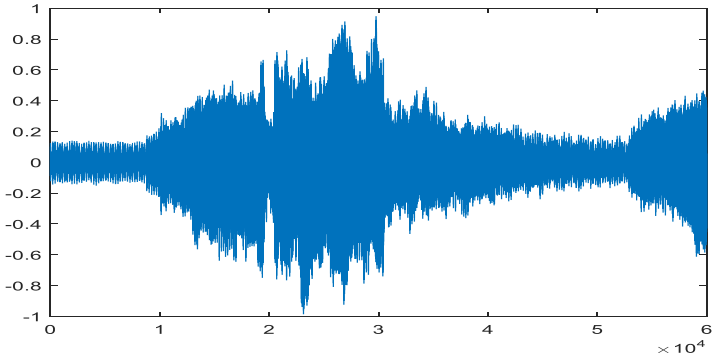
Application of Frequency Domain Analysis in Discrete Systems



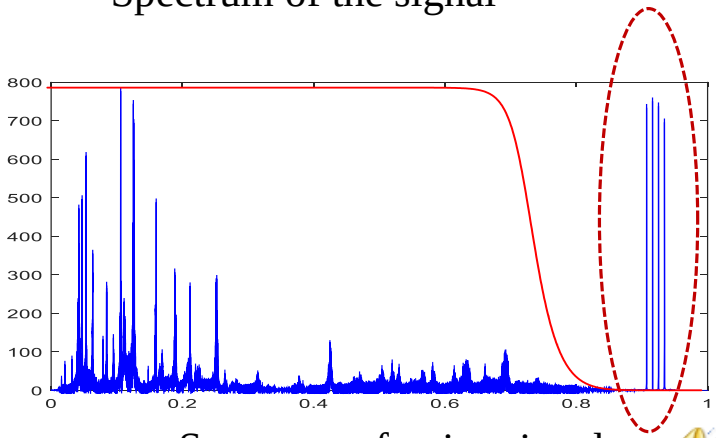
Time domain waveform of the signal



Spectrum of the signal



Time domain waveform of noisy signal



Spectrum of noisy signals

Filter out noise in the signal

[E.g] Given $-\pi \le \Omega \le \pi$ the frequency response $H(\mathrm{e}^{\mathrm{j}\Omega})$ of a discrete LTI system is

$$H(\mathrm{e}^{\mathrm{j}\Omega}) = \begin{cases} \mathrm{e}^{-\mathrm{j}3\Omega}, & |\Omega| < \dfrac{3\pi}{16} \\ 0, & \text{其他} \end{cases}$$

(1) Calculate the unit impulse response $h[k]$ of the system.

(2) When the input of the system is $x[k] = \tilde{\delta}_{16}[k] = \sum_{n=-\infty}^{\infty} \delta[k-16n]$, find the output $y[k]$ of the system.

Solution:

$$\frac{\Omega_c}{\pi}\mathrm{Sa}(\Omega_c k) \qquad h[k] = \frac{3}{16}\mathrm{Sa}\left[\frac{3\pi}{16}(k-3)\right]$$

[E.g] Given $-\pi \le \Omega \le \pi$ the frequency response $H(e^{j\Omega})$ of a discrete LTI system is

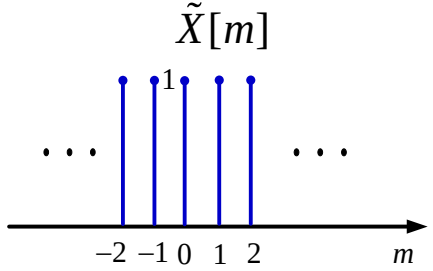
$$H(e^{j\Omega}) = \begin{cases} e^{-j3\Omega}, & |\Omega| < \dfrac{3\pi}{16} \\ 0, & \text{其他} \end{cases}$$

(1) Calculate the unit impulse response $h[k]$ of the system.

(2) When the input of the system is $x[k] = \tilde{\delta}_{16}[k] = \sum_{n=-\infty}^{\infty} \delta[k-16n]$, find the output $y[k]$ of the system.

Solution: $DFS\{\tilde{\delta}_{16}[k]\}$

$$\tilde{X}[m] = \sum_{k=0}^{N-1} \tilde{x}[k] e^{-j\Omega_0 mk}$$



[E.g] Given $-\pi \le \Omega \le \pi$ the frequency response $H(e^{j\Omega})$ of a discrete LTI system is

$$H(e^{j\Omega}) = \begin{cases} e^{-j3\Omega}, & |\Omega| < \dfrac{3\pi}{16} \\ 0, & \text{其他} \end{cases}$$

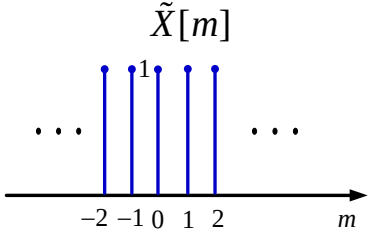
(1) Calculate the unit impulse response $h[k]$ of the system.

(2) When the input of the system is $x[k] = \tilde{\delta}_{16}[k] = \sum_{n=-\infty}^{\infty} \delta[k-16n]$, find the output $y[k]$ of the system.

Solution: Express $x[k]$ as a virtual exponential form

$$\tilde{x}[k] = \frac{1}{N}\sum_{m=0}^{N-1} \tilde{X}[m] e^{j\Omega_0 mk}$$

$$\tilde{\delta}_{16}[k] = \frac{1}{16}\sum_{m=-7}^{8} e^{j\frac{2\pi m}{16}k}$$



[E.g] Given $-\pi \le \Omega \le \pi$ the frequency response $H(\mathrm{e}^{\mathrm{j}\Omega})$ of a discrete LTI system is

$$H(\mathrm{e}^{\mathrm{j}\Omega}) = \begin{cases} \mathrm{e}^{-\mathrm{j}3\Omega}, & |\Omega| < \dfrac{3\pi}{16} \\ 0, & \text{其他} \end{cases}$$

(1) Calculate the unit impulse response $h[k]$ of the system.

(2) When the input of the system is $x[k] = \tilde{\delta}_{16}[k] = \sum_{n=-\infty}^{\infty} \delta[k-16n]$, find the output $y[k]$ of the system.

Solution:

$$\tilde{\delta}_{16}[k] = \frac{1}{16} \sum_{m=-7}^{8} \mathrm{e}^{\mathrm{j}\frac{2\pi m}{16}k}$$

$$y[k] = \frac{1}{16}\left\{ \mathrm{e}^{-\mathrm{j}\frac{2\pi}{16}k} \mathrm{e}^{\mathrm{j}3\frac{2\pi}{16}} + 1 + \mathrm{e}^{\mathrm{j}\frac{2\pi}{16}k} \mathrm{e}^{-\mathrm{j}3\frac{2\pi}{16}} \right\} = \frac{1}{16}\left\{ 1 + 2\cos[\frac{\pi}{8}(k-3)] \right\}$$

Thanks

The materials referenced in this course are accumulated over years of teaching by the instructor, sourced from various media and exchanges with colleagues and peers. It is difficult to indicate the source of each material one by one. This is to clarify and express gratitude.